

What is Seaborn?

Seaborn is a Python data visualization library based on Matplotlib. It provides a high-level interface for drawing attractive and informative statistical graphics.

Installation



```
pip install seaborn
```

Basic Plotting with Seaborn

First, let's import the necessary **libraries** and load a sample **dataset**.



```
In [1]: 1 import seaborn as sns
        2 import matplotlib.pyplot as plt
        3
        4 # Load a sample dataset
        5 tips = sns.load_dataset('tips')
```

```
In [2]: 1 tips.head(3)
```

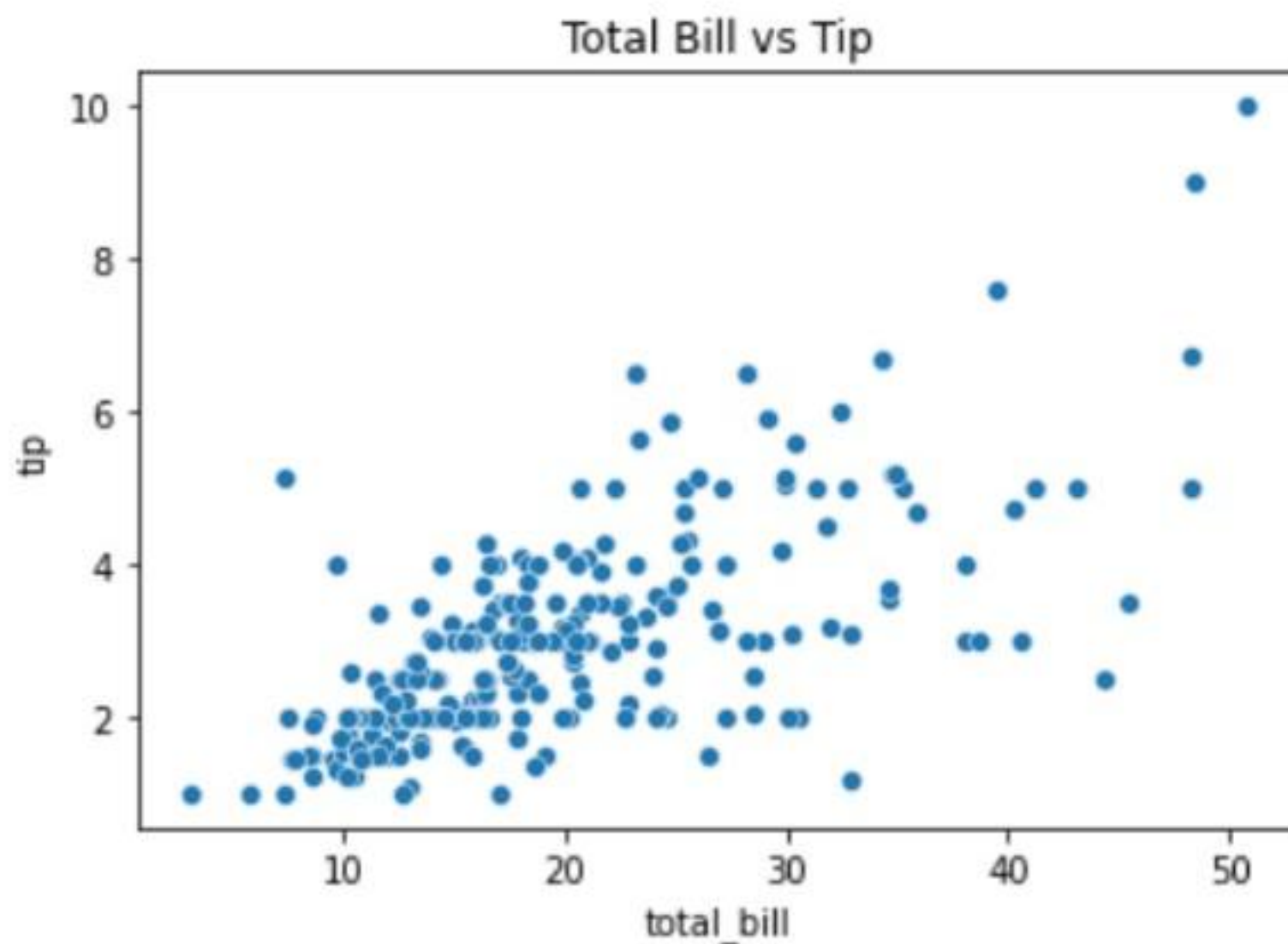
Out[2]:

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3

Scatter Plot

A scatter plot displays values for two variables as a collection of points.

```
In [3]: 1 # Scatter plot
        2 sns.scatterplot(data=tips, x='total_bill', y='tip')
        3 plt.title('Total Bill vs Tip')
        4 plt.show()
```

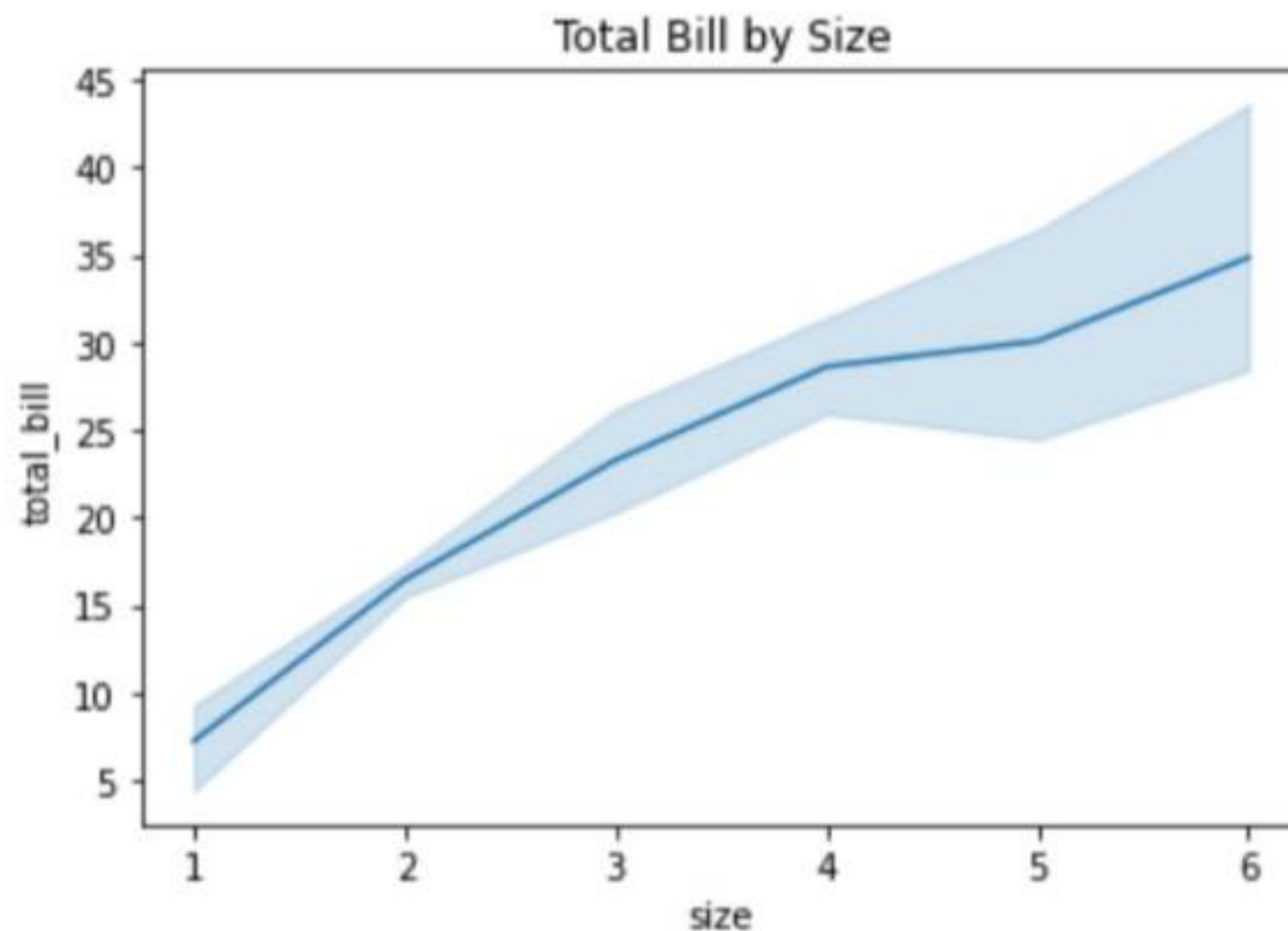


Line Plot

A line plot displays information as a series of data points connected by straight line segments.

In [4]:

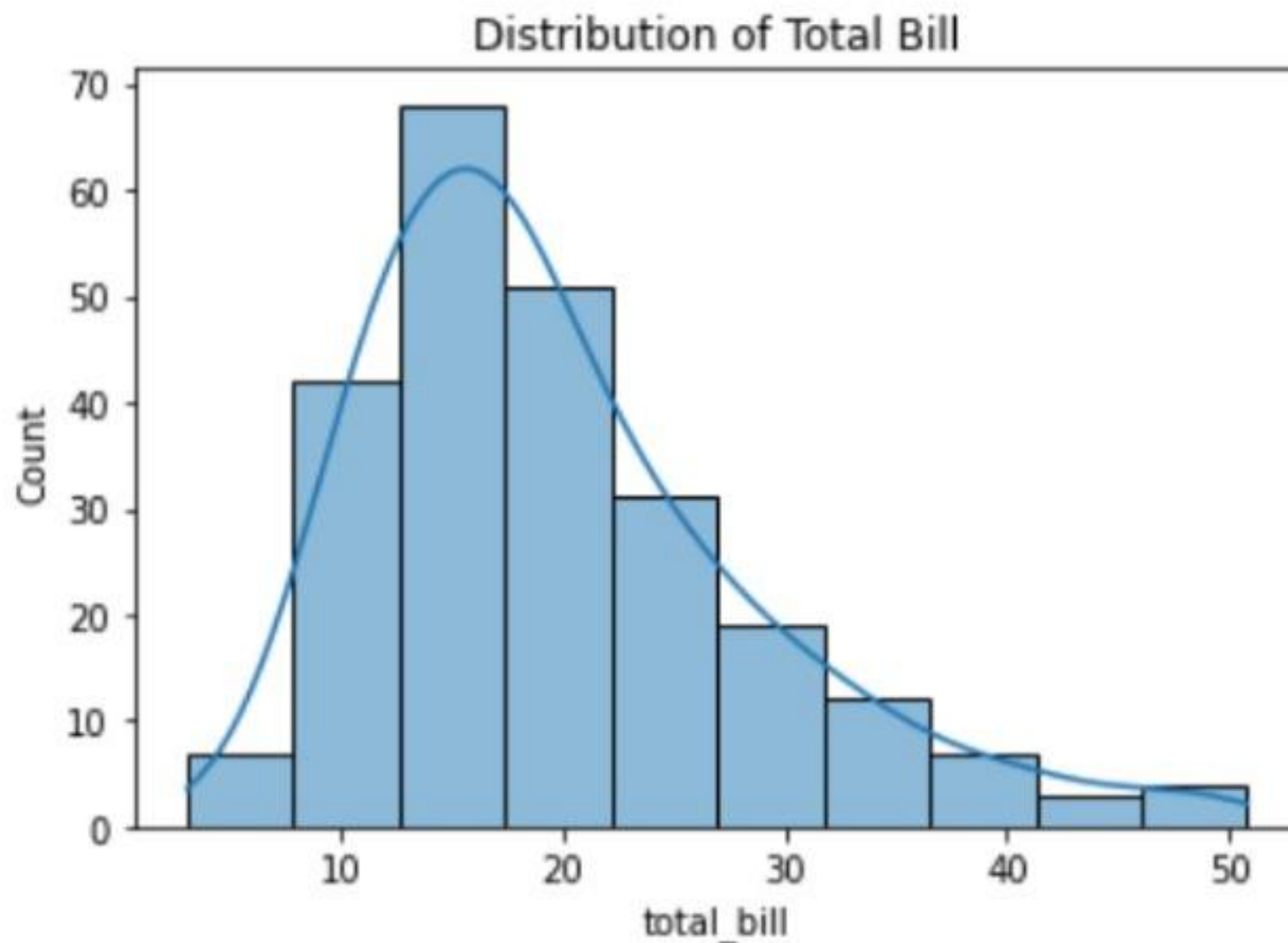
```
1 # Line plot
2 sns.lineplot(data=tips, x='size', y='total_bill')
3 plt.title('Total Bill by Size')
4 plt.show()
```



Histogram

A histogram is an accurate representation of the distribution of numerical data.

```
In [5]: 1 # Histogram
        2 sns.histplot(data=tips, x='total_bill', bins=10, kde=True)
        3 plt.title('Distribution of Total Bill')
        4 plt.show()
```



Bar Plot

A bar plot is a way of summarizing a set of categorical data.

In [6]:

```
1 # Bar plot
2 sns.barplot(data=tips, x='day', y='total_bill')
3 plt.title('Average Total Bill by Day')
4 plt.show()
```

